

HW3 – Knowledge Representation and Reasoning

(Propositional Logic, CNF, First-Order Logic)

Instructions

- Show your reasoning clearly for each question.
 - You may use any standard logical equivalences.
 - Unless otherwise stated, you can assume all variables range over an appropriate domain (e.g., people, courses, animals).
 - Write neatly and justify non-trivial steps; unreadable work may not receive full credit.
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1) Propositional Reasoning in a Tiny Wumpus World (30 pts)

We consider a simplified Wumpus world with a **2x2 grid**. The agent only cares about avoiding **pits**. The coordinates are (1, 1), (1, 2), (2, 1), (2, 2). The agent starts at square (1, 1).

We use the following propositional symbols:

- B_{ij} : There is a breeze in square (i, j) .
- P_{ij} : There is a pit in square (i, j) .

We assume the usual Wumpus-world rule for breezes and pits. In particular:

$$B_{11} \leftrightarrow (P_{12} \vee P_{21}).$$

In summary, the knowledge base KB for this question consists of:

- (i) There is breeze in (1,1)
- (ii) $B_{11} \leftrightarrow (P_{12} \vee P_{21})$.
- (iii) No other information about pits or breezes is known.

We are interested in entailment about $\alpha := P_{12}$.

(a) Truth Table (10 pts)

Fill a truth table involving propositions $B_{11}, P_{11}, P_{12}, P_{21}, P_{22}$ and KB .

B_{11}	P_{11}	P_{12}	P_{21}	P_{22}	KB

(b) Does KB entail P_{12} ? (10 pts)

Circle **one**:

$\text{KB} \models P_{12}$ $\text{KB} \not\models P_{12}$

Explain your answer in one line, by referring to the truth table.

(c) Does KB entail $\neg P_{12}$? (10 pts)

Circle **one**:

$\text{KB} \models \neg P_{12}$ $\text{KB} \not\models \neg P_{12}$

Explain your answer in one line, by referring to the truth table.

2) Conjunctive Normal Form (CNF) in the Wumpus World (30 pts)

(a) Propositional sentence (5 pts)

Consider the following English sentence:

“If there is no breeze in (1,1), then there is no pit in (1,2) and no pit in (2,1).”

Write this sentence as a propositional formula using B_{11} , P_{12} , and P_{21} .

(b) CNF conversion (25 pts)

Take your formula from part (a) and convert it to **equivalent CNF**. You should:

- Show each transformation step.
- Briefly name or describe the rule used at that step (e.g., “eliminate implication”, “push $\neg$ inward (De Morgan)”, “distribute $\vee$ over $\wedge$ ”, etc.).

3) First–Order Logic (FOL) Representation (40 pts)

Represent the following sentences in **First-Order Logic**. Use the most reasonable meaning for each predicate.

You may use any of the following predicates (you do not have to use all of them):

- $\text{Enrolled}(x, \text{CS})$: x is enrolled in the CS program.
- $\text{Takes}(x, \text{course})$: x takes course.
- $\text{Greedy}(x)$: x is greedy.
- $\text{Likes}(x, y)$: x likes y .
- $\text{Person}(x)$: x is a person.
- $\text{Hardworking}(x)$: x is hard-working.
- $\text{Takes}(x, y)$: x takes course y .
- $\text{Pass}(x, y)$: x passes course y .
- $\text{Halfsibling}(x, y)$: x and y are half-siblings.
- $\text{Father}(x, y)$: x is father of y .
- $\text{Mother}(x, y)$: x is mother of y .

(a) 10 pts

“Every CS student takes CS404 or CS408.”

(b) 10 pts

“No one likes greedy people.”

(c) 10 pts

“If a hard-working student will pass a course s/he is taking.”

(d) 10 pts

“Half siblings have a different mom or dad”